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Week 5: Image Processing

Week 5 Quiz

Quiz: Week 5 Image Processing

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Grade received 88.23%

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To pass 70% or greater

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Week 5 Image Processing I

1. Which of the following represents the gamma correction?

COOCH

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☒ $I_d(x, y) = I_r^{\gamma}(x, y) + \log I_r(x, y)$

☐ $I_d(x, y) = I_r(x^{\gamma})$

☐ $I_d(x, y) = I_r(\exp(x, y))$

☐ $I_d(x, y) = I_r(\log(x, \log(y)))$

Submit your assignment

Attempts: 3 every 1 hour

Retake the quiz in 7h 00m

Try again

Correct

Discrete grade: Pixel processing is changing the intensity at each pixel based solely on the current value of the intensity at the pixel. Pass 70% or higher

Your grade: 88.23%

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2. Which of the following is the expression for the convolution of two functions?

☐ $g(x) = \int_{-\infty}^{\infty} f(r)h(x-r)dr$

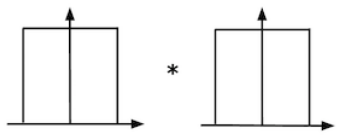
☐ $g(x) = \int_{-\infty}^{\infty} f(r)h(r-x)dr$

☐ $g(x) = f(r) \int_{-\infty}^{\infty} h(x-r)dr$

☐ $g(x) = \int_{-\infty}^{\infty} f(r)h(x-r)dr$

Correct

See definition of convolution.



3. Which of these is the result of the convolution of the two 1D signals shown?

☐

☐

☒

☐

Correct

Sketch out the solution.

4. Given two images, A and B, suppose that after convolving each image with function f we get a circle and a square, respectively, centered at the origin. Suppose we shift A to the left and B to the right and overlay the two images to form image C. What is the result of convolving image C with f?

☐ A circle to the right of a square

☒ A circle to the left of a square

☐ A circle and square centered at the origin

☐ None of the above

Correct

Convolution is a LSI which means that adding, shifting or scaling inputs before a convolution is equivalent to applying those same transformations after convolution.

5. There exists a convolution which can flip an image about its horizontal axis.

☐ True

☒ False

Correct

Convolutions must be shift invariant. Shifting all the pixels of an image up and then flipping is not equivalent to flipping an image and then shifting it up by the same amount.

6. You can find the impulse response (point blur function) of a camera by showing it:

☐ The sun

☐ The moon

☐ The sky

☒ A distant star

Correct

A distant bright star is equivalent to an impulse function, which is infinitesimally small and very bright (like a delta function). The image of the star directly yields the impulse response.

7. Which of the following is not a property of convolution?

☐ $u(x) * h(x) = h(x) * u(x)$

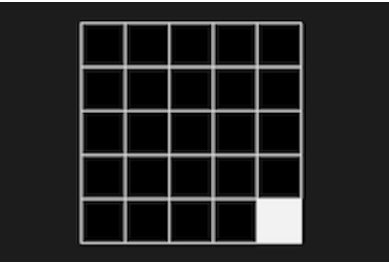
☐ $(u(x) * h(x)) * c(x) = u(x) * (h(x) * c(x))$

☐ $u(x) * h(x) + c(x) * c(x) = u(x) * (h(x) + c(x))$

☒ $u(x) * h(x) = u(x) * h(x)$

Correct

The first answer choice is the commutative property, the second answer choice is the associative property, and the third answer choice is the distributive property of convolution. The last answer choice is not valid.



8. The convolution of an image with the mask shown above will:

☐ Not affect the image

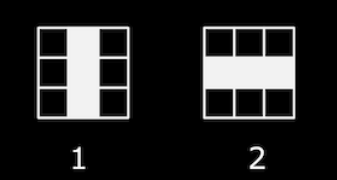
☐ Shift the image left and up by 2 pixels each

☐ Shift the image right and down by 2 pixels each

☐ Brighten the image

Correct

See how convolution works. Note that the mask will be flipped vertically and horizontally.



9. Convolving an image with the masks 1 and 2 in succession is equivalent to convolving the image with the following single mask:

☐

☐

☒

☐

Incorrect

10. Median filtering for noise removal can be implemented as:

☐ Convolution

☐ Correlation

☐ Gaussian smoothing

☒ None of the above

Correct

Median filtering is a non-linear operation, since it involves selecting a median (a non-linear operation). All the other operations listed are all linear.

11. Which of the following is the expression for 2D discrete correlation?

☐ $R_{fL}[i, j] = \sum_m \sum_n f[m, n]g[i-m, j-n]$

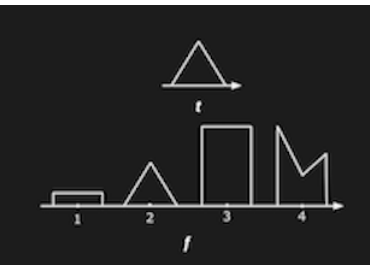
☒ $R_{fL}[i, j] = \sum_m \sum_n f[m, n]g[m-i, n-j]$

☐ $R_{fL}[i, j] = \sum_m \sum_n f[m, n] + g[m-i, n-j]$

☐ $R_{fL}[i, j] = \sum_m \sum_n f[m-i, n-j]$

Correct

See definition of 2D discrete correlation.



12. The simple correlation $R_{fL}[i, j]$ between the template L and the signal f would produce a maximum value at:

☐ 1

☐ 2

☒ 3

☐ 4

Correct

At 3 the value of the integral of the product of the template L and the signal f would be maximized.